

NEXUS Architecture

NEURAL EXCHANGE FOR UNIFIED STRUCTURED / UNSTRUCTURED INTELLIGENCE

FRONTEND

STREAMLIT UI

Conversational Chat Interface

st.chat_input

st.chat_message

st.session_state (msg history)

st.write_stream (LLM streaming)

st.cache_resource (Chroma + Embedder + Engine)

HITL approval widget

Unique session ID → SQLiteSession key → shared across all agent turns

user NLP query

GUARD

INPUTGUARDRAIL

run_in_parallel=False (runs BEFORE orchestrator)

SafetyGuardrailAgent

Safety check · Scope validation · Prompt injection detection

model: gpt-4o-mini → claude-haiku-3 fallback

PASS → Orchestrator proceeds

BLOCK → InputGuardrailTripwireTriggered → user sees reason

safe + in-scope query

ORCHESTRATOR

AGENT · HUB

SQLiteSession (persistent memory)

RunHooks: NEXUSRunHooks

NEXUSOrchestrator

Routes queries · Coordinates tool calls · Decides handoff to AnswerSynthesisAgent

PRIMARY: claude-sonnet-4 (LitellmModel) | FALLBACK: gpt-4o

ROUTING LOGIC

Employee query only

News / knowledge query

Employee + weather

Data refresh needed

Multi-source synthesis

→ call **sql_tool**

→ call **rag_retriever_tool**

→ sql_tool → extract office_location → **weather_fusion_tool** → handoff

→ call **ingestion_tool**

→ **HANDOFF** to AnswerSynthesisAgent

as_tool() calls

HANDOFF (remove_all_tools filter)

🔧 TOOL TIER — AGENTS EXPOSED VIA .AS_TOOL()

AGENT → TOOL

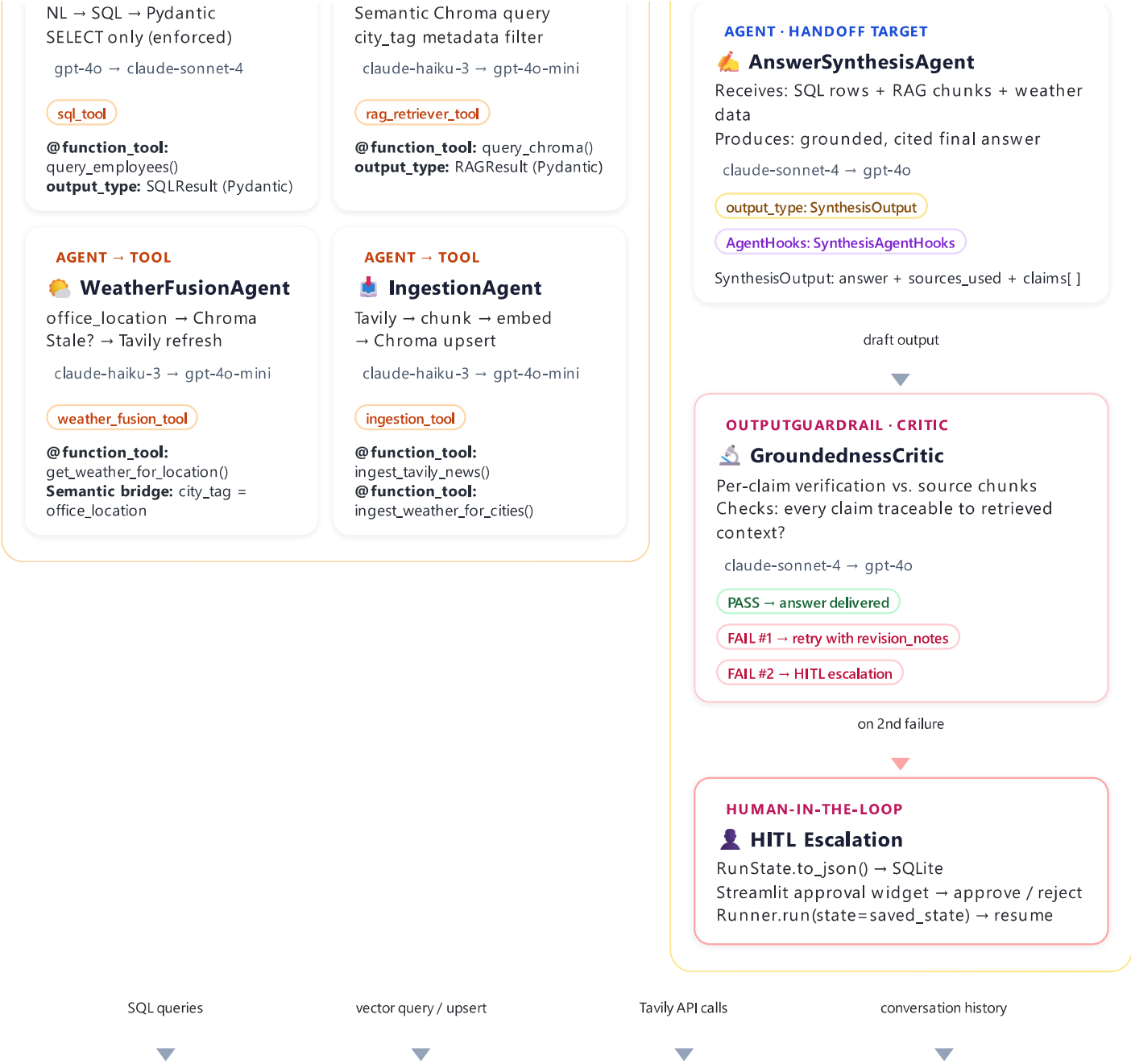
SQLAgent

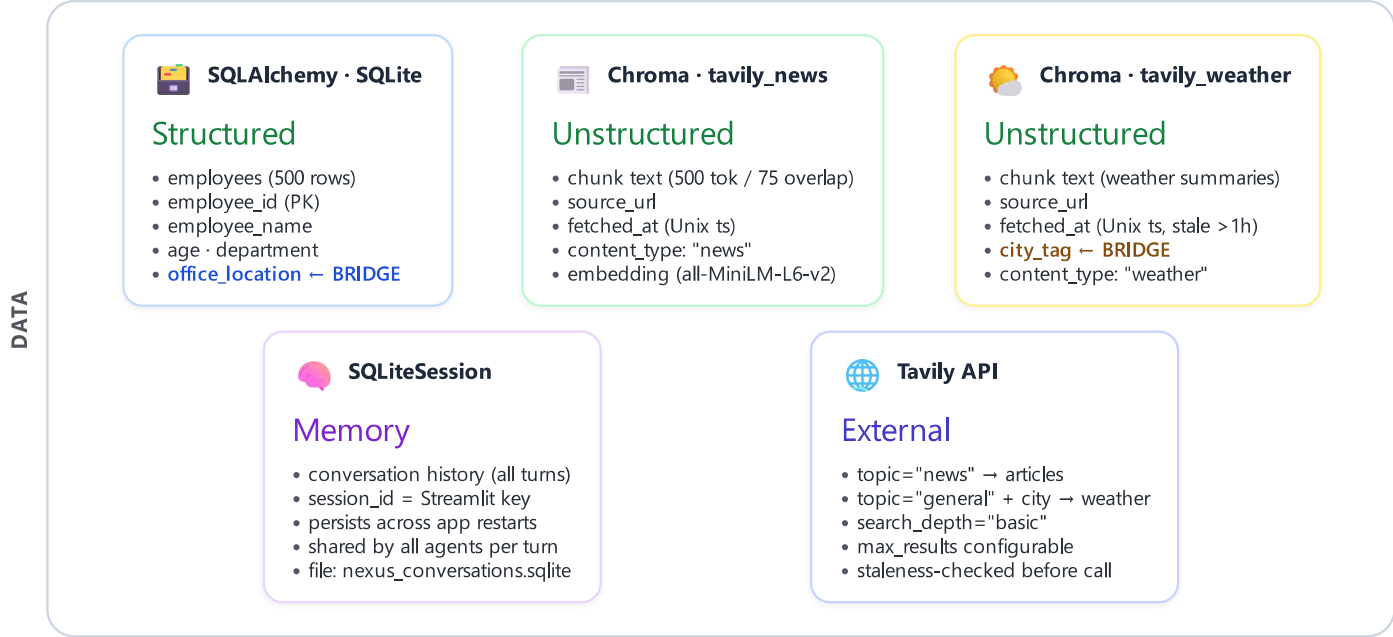
AGENT → TOOL

RAGRetrieverAgent

🔥 SYNTHESIS + GROUNDEDNESS

file:///C:/Users/vmuser/AppData/Local/Claude-3p/local-agent-mode-sessions/3a2fba0f-3bfd-4471-9b86-0d6ec5f7e911/00000000-0000-4000-8000-... 1/3







THE SEMANTIC BRIDGE — How Structured Meets Unstructured

employees.office_location (SQL, e.g. "Austin") = chroma.tavily_weather[city_tag] (Chroma metadata, e.g. "Austin")

WeatherFusionAgent queries Chroma with where={"city_tag": {"\$eq": office_location}}

Contract: CITIES list in seed.py must exactly match city_tags ingested by IngestionAgent.

OPENAI AGENTS SDK PRIMITIVES USED

Agent

Runner.run_streamed()

RunConfig

RunHooks

AgentHooks

@function_tool

Agent.as_tool()

handoff()

handoff(input_filter=remove_all_tools)

@input_guardrail

@output_guardrail

GuardrailFunctionOutput

InputGuardrailTripwireTriggered

OutputGuardrailTripwireTriggered

SQLiteSession

RunState.to_json() / from_json()

LitellmModel

LitellmProvider

RunContextWrapper

output_type (Pydantic)

☐ Orchestrator / Functional Agent

☐ Agent exposed as Tool (as_tool)

☐ Critic / Guardrail Agent

☐ Data Store

☐ Memory / Session

☐ External API

→ Normal flow

---> Handoff

 Semantic Bridge: office_location = city_tag